

Feature	Cluster	Model	Acc.	Prec.	Recall	F1
gender	Male	GP-FL	0.9532	0.8423	0.6403	0.7275
	Female	GP-FL	0.9675	0.8864	0.6412	0.7441
age	Cluster 0 (0.1–40.0)	GP-FL	0.9903	0.9242	0.4150	0.5728
	Cluster 1 (41.0–80.0)	GP-FL	0.9346	0.8397	0.6745	0.7481
hypertension	No Hypertension	GP-FL	0.9680	0.8556	0.6342	0.7284
	Hypertension	GP-FL	0.8831	0.8433	0.7115	0.7718
heart_disease	No Heart Disease	GP-FL	0.9664	0.8837	0.6322	0.7371
	Heart Disease	GP-FL	0.8669	0.8286	0.7160	0.7682
smoking_history	never	GP-FL	0.9530	0.7532	0.7227	0.7377
	No Info	GP-FL	0.9800	0.8267	0.6073	0.7002
	current	GP-FL	0.9510	0.8012	0.6963	0.7451
	former	GP-FL	0.9236	0.8622	0.6697	0.7539
	ever	GP-FL	0.9363	0.7327	0.7551	0.7437
	not current	GP-FL	0.9488	0.8230	0.6691	0.7381
bmi	Underweight (< 18.5)	GP-FL	0.9947	1.0000	0.3077	0.4706
	Healthy (18.5–24.9)	GP-FL	0.9802	0.8264	0.5988	0.6944
	Overweight (25–29.9)	GP-FL	0.9644	0.8343	0.6286	0.7170
	Obesity I (30–34.9)	GP-FL	0.9297	0.8119	0.6685	0.7332
	Obesity II (35–39.9)	GP-FL	0.9045	0.7811	0.7309	0.7552
	Obesity III (≥ 40)	GP-FL	0.8865	0.7566	0.7738	0.7651
HbA1c_level	Normal (< 5.7)	GP-FL	1.0000	0.0000	0.0000	0.0000
	Pre-diabetes (5.7–6.4)	GP-FL	0.9375	0.6506	0.5176	0.5766
	Diabetes (≥ 6.5)	GP-FL	0.9454	0.9342	0.8264	0.8770
blood_glucose_level	Normal (70–99)	GP-FL	1.0000	0.0000	0.0000	0.0000
	Pre-diabetes (100–125)	GP-FL	1.0000	0.0000	0.0000	0.0000
	Diabetes (≥ 126)	GP-FL	0.9475	0.8267	0.6902	0.7523

Table 1: GP-FL performance for Dataset 1